OpenGonio wiki

Explanations, bibliography (APA 7) and background appendices

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OpenGonio wiki

Open-source four-axis gonioreflectometer and 3D scanner. It measures the bidirectional reflectance distribution function (BRDF) of flat samples and scans objects in 3D. Design follows Ansari-Asl et al. (2024), rebuilt for under $630 in parts, with no soldering, drilling, sawing or crimping.

Page	Contents
How it works	BRDF, geometry, the four axes, one measurement
Hardware	subsystems, joins without tools, cost
Calibration	angles, camera response, absolute reflectance, drift

Page	Contents
3D scanning	fixed camera, visual hull, photogrammetry, photometric stereo
Software	`opengonio` commands, simulator, data
Paper	manuscript for *HardwareX*, build, status
Bibliography	every scholarly source, APA 7
Appendices A · B · C · D · E	background papers by topic, with what each is used for

Build guide, bill of materials and commands: README. Step checklists: phases.

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How it works

What is measured

The BRDF f_r is reflected radiance per unit incident irradiance, for one incoming and one outgoing direction (Hanrahan, 2016; Nicodemus et al., 1977):

$$f_r(\omega_i \to \omega_r) = \frac{dL_r(\omega_i \to \omega_r)}{dE_i} \; [\mathrm{sr}^{-1}]$$

Isotropic materials need 3 angles; anisotropic ones (brushed metal, satin) need 4 (Ward, 1992).

Geometry

One zero point; every axis turns about it.

Axis	Moves	Range
M4	light arm around the sample	20–324°
M3	table (sample yaw)	±180°
M2	sample tilt	±90°
M1	sample spin	±180°

- Measuring camera fixed at 350 mm; light 300 mm from the zero point.
- Light on a horizontal ring: M3 and M2 already turn the sample to every direction an overhead light could reach, so the arm stays low, stiff and out of the light path (Ansari-Asl et al., 2024).
- In-plane, the mirror peak sits at M3 = M4 / 2.
- Light within 20° of the camera is out of reach: pure back-scatter is not measured.

One measurement

1. Move to a pose (inverse kinematics from $\theta_i, \theta_r, \Delta\varphi$).
2. Capture at fixed exposure; several exposures for high dynamic range (HDR) (Debevec & Malik, 1997).
3. Subtract dark frame, linearise, divide by the white reference (Calibration).
4. Fit Lambert + GGX microfacet model (Trowbridge & Reitz, 1975; Walter et al., 2007) → game-engine material.

Full equations: README theory.

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Hardware

Subsystem	Parts	Stage
Power	24 V 60 W adapter, fuse, latching e-stop, lever connectors	01
Controller	FYSETC E4 (FluidNC) + Raspberry Pi Pico H, keypads	02
Frame	610 × 610 mm plywood, 2020 extrusion, printed mounts	03
Light arm (M4)	12 in lazy Susan, GT2 belt rim, 24:1	04
Table (M3), tilt (M2), spin (M1)	NEMA 17 steppers, 5:1 belts, M8 bolt axles, direct-drive spin	05 · 08 · 09
Camera	12 MP IMX477 USB, 16 mm M12 lens, printed f/8 stop	06
Light	Citizen CLU7A3 COB, colour rendering index (CRI) 97, in a solderless TE holder, 350 mA driver	06

- **No tools beyond a kit:** lever connectors, screw terminals, breadboard, self-tapping screws into printed parts, wood screws without pilot holes.
- **Printed parts:** black ABS (acrylonitrile butadiene styrene) on an enclosed printer; ≈ 1 kg.
- **Optical homing** on every axis; over-travel switches in the e-stop loop.
- **Cost:** see the bill of materials (pictures, buy links, STEP/STL downloads).

Open-hardware practice (Open Source Hardware Association, n.d.; Pearce, 2012): Appendix E.

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Open Source Hardware Association. (n.d.). *Open source hardware (OSHW) definition 1.0*. Retrieved https://oshwa.org/definition/

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Calibration

Quantity	Method	Pass
Angle offset	mirror peak must sit at M3 = M4 / 2 (Ansari-Asl et al., 2024)	residual ≤ 0.1°
Dark level C_d	lamp-off frame	—
Camera response γ	exposure series (Debevec & Malik, 1997)	linear fit
Absolute BRDF	PTFE (polytetrafluoroethylene) tile, ≈ 99 % diffuse, at 45° (Weidner & Hsia, 1981)	—
Lamp drift	repeat the first pose at the end	≤ 2 %

$$f_r = \frac{C_{\text{lin}}}{C_{w,\text{lin}}} \cdot \frac{\rho_w}{\pi} \cdot \frac{\cos 45^\circ}{\cos \theta_i}, \qquad C_{\text{lin}} = (C - C_d)^\gamma$$

C camera counts, C_w white tile, ρ_w its reflectance. Steps: stage 07.

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3D scanning

The camera never moves; the sample axes turn the object, so every photo's pose is known.

1. **Pose from the axes:** camera calibrated once against the turntable (hand-eye, AX = YB) (Shah, 2013).
2. **Quick shape:** silhouettes → visual hull (Laurentini, 1994).
3. **Full mesh:** photogrammetry (Griwodz et al., 2021), then screened Poisson reconstruction (Kazhdan & Hoppe, 2013).
4. **Surface detail (optional LED bank):** photometric stereo from known light directions (Ackermann & Goesele, 2015).
5. **Material:** measured BRDF fitted and textured onto the mesh; export to a game engine.

Stages 11 · 12 · 13. Background: Appendix C.

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Software

Python package `opengonio` (software/). Motion: FluidNC on the E4; the Pico reads keypads and sensors.

```
make setup && source .venv/bin/activate
make sim                                    # simulated machine, ends "ALL PASS"
opengonio check-pose M4=60 M3=30            # angles + collision check
opengonio --sim plastic capture inplane --step 10 --out data/sim/plastic
opengonio curve data/sim/plastic            # reflectance curve
```

- Laptop mode: the laptop runs jobs. Standalone mode: a Pi Zero 2 W runs them; `opengonio pull` + `opengonio verify` bring data back.
- `make all`: rebuild docs, checks, tests.

Paper

Manuscript for *HardwareX* (Elsevier), the open-source scientific hardware journal: paper/.

- PDF: paper/build/opengonio.pdf, rebuilt by `make paper` on every change.
- Sections follow the HardwareX hardware-article template; citations APA 7 (`biblatex-apa`), keys = reference IDs.
- Edit elsewhere: Word · Overleaf zip · references for Zotero; copy edits back into `paper/opengonio.tex`.
- Results sections wait for measured data (stages 06–09).

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Appendix A · Reflectance measurement and gonioreflectometers

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

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Lineage of the main design reference (NTNU / LNE-CNAM)

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- Filip, J., Vávra, R., & Krupička, M. (2014). Rapid material appearance acquisition using consumer hardware. *Sensors*, *14*(10), 19785–19805. https://doi.org/10.3390/s141019785 — *Toy-gantry low-cost BTF prior art (phases 00, 03)* [G38]
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- Riviere, J., Peers, P., & Ghosh, A. (2016). Mobile surface reflectometry. *Computer Graphics Forum*, *35*, 191–202. https://doi.org/10.1111/cgf.12719 — *Phone-flash SVBRDF capture baseline (phase 09)* [G40]

Spherical, polarized and linear-light illumination; SVBRDF capture

- Ghosh, A., Chen, T., Peers, P., Wilson, C. A., & Debevec, P. (2009). Estimating specular roughness and anisotropy from second order spherical gradient illumination. *Eurographics Symposium on Rendering*. https://www.cs.wm.edu/~ppeers/publications/Ghosh2009ESR/Ghosh_EGSR2009.pdf — *Polarized diffuse/specular separation (phase 10)* [G41]
- Ghosh, A., Chen, T., Peers, P., Wilson, C. A., & Debevec, P. (2010). Circularly polarized spherical illumination reflectometry. *ACM SIGGRAPH Asia*. https://vgl.ict.usc.edu/Research/CPSIR/ —

Polarization-based index/specular estimation (phase 10) [G42]
- Chen, G., Dong, Y., Peers, P., Zhang, J., & Tong, X. (2014). Reflectance scanning: Estimating shading frame and BRDF with generalized linear light sources. *ACM Transactions on Graphics*, *33*(4). https://doi.org/10.1145/2601097.2601180 — *Linear-light SVBRDF scanning (phase 09)* [G43]
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- Baek, S.-H., Zeltner, T., Ku, H., Hwang, I., Tong, X., Jakob, W., & Kim, M. H. (2020). Image-based acquisition and modeling of polarimetric reflectance. *ACM Transactions on Graphics*, *39*(4), 139. https://doi.org/10.1145/3386569.3392387 — *Polarimetric BRDF capture (phase 10)* [G45]

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- Vávra, R., & Filip, J. (2016). Minimal sampling for effective acquisition of anisotropic BRDFs. *Computer Graphics Forum*, *35*(7), 299–309. https://www.pbr-book.org/4ed/Reflection_Models/Further_Rea — *Anisotropic sparse sampling (phase 17)* [G47]
- Vávra, R., & Filip, J. (2018). Adaptive slices for acquisition of anisotropic BRDF. *Computational Visual Media*, *4*(1), 55–69. https://doi.org/10.1007/s41095-017-0099-z — *Adaptive 1D-slice sampling (phase 17)* [G48]

BTF capture setups (Bonn)

- Schwartz, C., Sarlette, R., Weinmann, M., Rump, M., & Klein, R. (2014). Design and implementation of practical bidirectional texture function measurement devices focusing on the developments at the university of Bonn. *Sensors*, *14*(5), 7753–7819. https://doi.org/10.3390/s140507753 — *BTF device requirements, HDR, calibration (phase 09)* [G49]
- Schwartz, C., Sarlette, R., Weinmann, M., & Klein, R. (2013). DOME II: A parallelized BTF acquisition system. *Eurographics Workshop on Material Appearance Modeling*. https://doi.org/10.2312/MAM.MAM2013.025-031 — *Camera-array dome comparison (phase 09)* [G50]

Datasets and data formats

- Matusik, W., Pfister, H., Brand, M., & McMillan, L. (2023). *BRDF database – bidirectional reflectance distribution function*. https://doi.org/10.5281/zenodo.8101681 — *Licensed MERL copy for comparison (phase 17)* [G51]
- Lanevski, D. et al. (2022). Machine-readable universal data format for bidirectional reflectance distribution function and BiRDview—an open-source web-based application for viewing and comparing bidirectional reflectance data. *Color Research & Application*. https://doi.org/10.1002/col.22790 — *BRDF export format, viewer (phase 14)* [G52]

Appendix B · Reflectance theory and models

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

Rendering theory and textbooks

- Kajiya, J. T. (1986). The rendering equation. *Proc. SIGGRAPH '86, Computer Graphics*, *20*(4), 143–150. https://doi.org/10.1145/15922.15902 — *Reflection/rendering equation background (phases 00, 07)* [T1]
- Pharr, M., Jakob, W., & Humphreys, G. (2023). *Physically based rendering: From theory to implementation* (4th ed.). https://www.pbr-book.org/4ed/Reflection_Models/Further_Reading — *BRDF, microfacet, sampling reference (phases 07, 08)* [T2]
- Veach, E. (1997). *Robust Monte Carlo methods for light transport simulation*. https://graphics.stanford.edu bw.pdf — *Reciprocity, importance sampling theory (phases 08, 17)* [T3]
- Weyrich, T., Lawrence, J., Lensch, H. P. A., Rusinkiewicz, S., & Zickler, T. (2009). Principles of appearance acquisition and representation. *Foundations and Trends in Computer Graphics and Vision*, *4*(2), 75–191. https://doi.org/10.1561/0600000022 — *Measurement-to-representation overview (phases 00, 08, 09)* [T4]
- Montes, R., & Ureña, C. (2012). *An overview of BRDF models*. https://digibug.ugr.es/bitstream/handle/104 2012-001TR.pdf — *Model catalogue for choosing fits (phases 07, 08)* [T5]
- Nimier-David, M., Vicini, D., Zeltner, T., & Jakob, W. (2019). Mitsuba 2: A retargetable forward and inverse renderer. *ACM Transactions on Graphics*, *38*(6), 203. https://doi.org/10.1145/3355089.3356498 — *Differentiable reference renderer for fit checks (phase 17)* [T6]
- Mitsuba 3 documentation. (n.d.). *BSDFs*. https://mitsuba.readthedocs.io/en/stable/src/generated/plugins_ — *Render fitted vs RGL measured BSDF (phase 17)* [T7]

Classic analytic reflectance models

- Phong, B. T. (1975). Illumination for computer generated pictures. *Communications of the ACM*, *18*(6), 311–317. https://doi.org/10.1145/360825.360839 — *Baseline lobe model for comparisons (phase 07)* [T8]
- Blinn, J. F. (1977). Models of light reflection for computer synthesized pictures. *Proc. SIGGRAPH '77*. https://doi.org/10.1145/563858.563893 — *Half-vector lobe, legacy export mapping (phase 07)* [T9]
- Torrance, K. E., & Sparrow, E. M. (1967). Theory for off-specular reflection from roughened surfaces. *Journal of the Optical Society of America*, *57*(9), 1105–1114. https://doi.org/10.1364/JOSA.57.00110 — *Microfacet theory, off-specular peak (phase 08)* [T10]
- Trowbridge, T. S., & Reitz, K. P. (1975). Average irregularity representation of a rough surface for ray reflection. *Journal of the Optical Society of America*, *65*(5), 531–536. https://doi.org/10.1364/JOSA.65.000531 — *Origin of GGX/TR distribution (phase 07)* [T11]
- Smith, B. G. (1967). Geometrical shadowing of a random rough surface. *IEEE Transactions on Antennas and Propagation*, *15*(5), 668–671. https://ui.adsabs.harvard.edu/abs/1967ITAP...15..668S/abstra — *Smith masking-shadowing term (phases 07, 08)* [T12]
- Oren, M., & Nayar, S. K. (1994). Generalization of Lambert's reflectance model. *Proc. SIGGRAPH '94*. https://doi.org/10.1145/192161.192213 — *Rough-diffuse alternative to Lambert (phases 07, 08)* [T13]
- Lafortune, E. P. F., Foo, S.-C., Torrance, K. E., & Greenberg, D. P. (1997). Non-linear approximation of reflectance functions. *Proc. SIGGRAPH '97*. https://doi.org/10.1145/258734.258801 — *Multi-lobe fit, retroreflection (phase 08)* [T14]
- Ashikhmin, M., & Shirley, P. (2000). An anisotropic Phong BRDF model. *Journal of Graphics Tools*, *5*(2), 25–32. https://doi.org/10.1080/10867651.2000.10487522 — *Anisotropic fit alternative (phase 09)* [T15]
- Rusinkiewicz, S. (1998). A new change of variables for efficient BRDF representation. *Rendering Techniques '98*. https://doi.org/10.1007/978-3-7091-6453-2_2 — *Half/difference angle*

data layout (phases 08, 17) [T16]

Microfacet theory and sampling (modern)

- Heitz, E. (2014). Understanding the masking-shadowing function in microfacet-based BRDFs. *Journal of Computer Graphics Techniques*, *3*(2), 48–107. https://jcgt.org/published/0003/02/03/paper.pdf — *Correct Smith G for GGX fit (phases 07, 08)* [T17]
- Heitz, E., & d'Eon, E. (2014). Importance sampling microfacet-based BSDFs using the distribution of visible normals. *Computer Graphics Forum*, *33*(4), 103–112. https://doi.org/10.1111/cgf.12417 — *Sampling fitted GGX in renders (phases 14, 17)* [T18]
- Heitz, E. (2018). Sampling the GGX distribution of visible normals. *Journal of Computer Graphics Techniques*, *7*(4), 1–13. https://jcgt.org/published/0007/04/01/paper.pdf — *Simple VNDF sampler code (phase 14)* [T19]
- Dupuy, J., & Benyoub, A. (2023). Sampling visible GGX normals with spherical caps. *Computer Graphics Forum*, *42*(8). https://doi.org/10.1111/cgf.14867 — *Faster VNDF sampling (phase 14)* [T20]
- Heitz, E., Hanika, J., d'Eon, E., & Dachsbacher, C. (2016). Multiple-scattering microfacet BSDFs with the Smith model. *ACM Transactions on Graphics*, *35*(4), 58. https://doi.org/10.1145/2897824.292 — *Energy loss at high roughness (phase 08)* [T21]
- Turquin, E. (2019). *Practical multiple scattering compensation for microfacet models*. https://blog.selfshadow.com/publications/turquin/ms_comp_final.pdf — *Energy compensation when fitting rough metals (phase 07)* [T22]
- Bagher, M. M., Soler, C., & Holzschuch, N. (2012). Accurate fitting of measured reflectances using a shifted gamma micro-facet distribution. *Computer Graphics Forum*, *31*(4). https://inria.hal.science/hal-00702304 — *Long-tail NDF fitting option (phase 17)* [T23]
- Ribardière, M., Bringier, B., Meneveaux, D., & Simonot, L. (2017). STD: Student's t-distribution of slopes for microfacet based BSDFs. *Computer Graphics Forum*, *36*(2). https://doi.org/10.1111/cgf.13137 — *Generalised GGX tail fitting (phase 17)* [T24]

Fitting measured data and evaluating fits

- Löw, J., Kronander, J., Ynnerman, A., & Unger, J. (2012). BRDF models for accurate and efficient rendering of glossy surfaces. *ACM Transactions on Graphics*, *31*(1), 9. https://doi.org/10.1145/2077341.2077350 — *Fit error metric, glossy model choice (phases 07, 08)* [T25]
- Dupuy, J., Heitz, E., Iehl, J.-C., Poulin, P., & Ostromoukhov, V. (2015). Extracting microfacet-based BRDF parameters from arbitrary materials with power iterations. *Computer Graphics Forum*, *34*(4), 21–30. https://doi.org/10.1111/cgf.12675 — *Robust NDF extraction for GGX fit (phases 08, 09)* [T26]
- Lavoué, G., Bonneel, N., Farrugia, J.-P., & Soler, C. (2021). Perceptual quality of BRDF approximations: Dataset and metrics. *Computer Graphics Forum*, *40*(2). https://doi.org/10.1111/cgf.142636 — *Perceptual fit-quality metric (phases 07, 17)* [T27]
- Kavoosighafi, B. et al. (2025). Perceived quality of BRDF models. *Computer Graphics Forum*. https://doi.org/10.1111/cgf.70162 — *Which model looks best when fitted (phase 17)* [T28]
- Sztrajman, A., Rainer, G., Ritschel, T., & Weyrich, T. (2021). Neural BRDF representation and importance sampling. *Computer Graphics Forum*, *40*. https://doi.org/10.1111/cgf.14335 — *Neural fit stretch; remap to analytic (phase 17)* [T29]

Fresnel and optical constants

- Born, M., & Wolf, E. (2019). *Principles of optics* (7th ed.). https://www.cambridge.org/9781108477437 — *Exact Fresnel equations, polarisation (phases 07, 10)* [T30]
- Gulbrandsen, O. (2014). Artist friendly metallic Fresnel. *Journal of Computer Graphics Techniques*, *3*(4), 64–72. https://jcgt.org/published/0003/04/03/paper.pdf — *Map n,k to reflectivity/edge tint (phase 07)* [T31]

- Polyanskiy, M. N. (2024). Refractiveindex.info database of optical constants. *Scientific Data*, *11*, 94. https://doi.org/10.1038/s41597-023-02898-2 — *Metal F0 sanity check vs mirror, chrome (phases 06, 07)* [T32]

Production shading models (course notes)

- Burley, B. (2012). Physically-based shading at Disney. *Practical Physically Based Shading in Film and Game Production*. https://static1.squarespace.com/static/58586fa5ebbd1a60e7d76d3e/t/593a3a — *Principled parameters, MERL-based design (phases 07, 17)* [T33]
- Burley, B. (2015). *Extending the Disney BRDF to a BSDF with integrated subsurface scattering*. https://blog.selfshadow.com/publications/s2015-shading-course/burley/s2015_pbs_disney_bsdf_notes.pdf — *Anisotropic alpha mapping, sheen (phase 09)* [T34]
- Karis, B. (2013). Real shading in Unreal engine 4. *Physically Based Shading in Theory and Practice*. https://cdn2.unrealengine.com/Resources/files/2013SiggraphPresentationsNotes-26915738.pdf — *UE GGX/Smith/Schlick, α = roughness² (phase 07)* [T35]
- Lagarde, S., & Rousiers, C. de. (2014). *Moving Frostbite to physically based rendering 3.0*. https://seblagarde.wordpress.com/wp-content/uploads/2015/07/course_notes_moving_frostbite_to_pbr_v3 — *Validation against references, photometric units (phase 07)* [T36]
- Kulla, C., & Conty, A. (2017). *Revisiting physically based shading at Imageworks*. https://blog.selfshadow.com/publications/s2017-shading-course/imageworks/s2017_pbs_imageworks_slid — *Energy compensation, Charlie sheen (phases 07, 09)* [T37]
- Hoffman, N. (2015). Physics and math of shading. *Physically Based Shading in Theory and Practice*. https://blog.selfshadow.com/publications/s2015-shading-course/hoffman/s2015_pbs_physics_ma — *Plain-language theory for docs (phases 00, 07)* [T38]
- Guy, R., & Agopian, M. (n.d.). *Physically based rendering in Filament*. https://google.github.io/filament/Fila — *Readable real-time BRDF derivations (phases 07, 14)* [T39]

Material exchange formats and engine docs

- Khronos Group. (n.d.). *glTF 2.0 specification*. https://registry.khronos.org/glTF/specs/2.0/glTF-2.0.html#metallic-roughness-material — *Core export target and BRDF (phases 07, 13)* [T40]
- *Khronos, KHR_materials_anisotropy*. (n.d.). https://github.com/KhronosGroup/glTF/blob/main/extensions — *Export fitted anisotropy (phase 09)* [T41]
- *Khronos, KHR_materials_sheen, KHR_materials_clearcoat*. (n.d.). https://github.com/KhronosGroup/glTF/ — *Fabric and coated-sample export (phase 09)* [T42]
- *Khronos, KHR_materials_specular, KHR_materials_ior*. (n.d.). https://github.com/KhronosGroup/glTF/blob — *Export measured dielectric F0 (phase 07)* [T43]
- *Khronos, KHR_texture_transform*. (n.d.). https://github.com/KhronosGroup/glTF/blob/main/extensions/2.0 — *SVBRDF texture tiling on models (phases 09, 13)* [T44]
- *Khronos, glTF sample viewer*. (n.d.). https://github.com/KhronosGroup/glTF-Sample-Viewer — *Reference check of exported .glb (phases 07, 13)* [T45]
- Epic Games. (n.d.). *Physically based materials in Unreal engine*. https://dev.epicgames.com/documentation/en-us/unreal-engine/physically-based-materials-in-unreal-engine — *Unreal export mapping (phases 07, 13)* [T46]
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- Blender Manual. (n.d.). *Principled BSDF*. https://docs.blender.org/manual/en/latest/render/shader_nodes/ — *Check exports in Blender (phases 07, 13)* [T48]
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- *ASWF, MaterialX specification and physically based shading nodes*. (n.d.). https://github.com/AcademySof — *Renderer-neutral material file (phases 13, 16)* [T51]

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- Jakob, W., & Hanika, J. (2019). A low-dimensional function space for efficient spectral upsampling. *Computer Graphics Forum*, *38*(2), 147–155. https://doi.org/10.1111/cgf.13626 — *Smooth spectra from fitted RGB albedo (phase 10)* [T53]
- Peters, C., Merzbach, S., Hanika, J., & Dachsbacher, C. (2019). Using moments to represent bounded signals for spectral rendering. *ACM Transactions on Graphics*, *38*(4), 136. https://doi.org/10.1145/3306346.3322964 — *Compact storage of 8-band reflectance (phase 10)* [T54]
- Wilkie, A., Nawaz, S., Droske, M., Weidlich, A., & Hanika, J. (2014). Hero wavelength spectral sampling. *Computer Graphics Forum*, *33*(4), 123–131. https://doi.org/10.1111/cgf.12419 — *Spectral preview render of measured bands (phases 10, 14)* [T55]

Appendix C · Calibration, HDR, photometric stereo, 3D scanning

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

3D photo scanner

- *COLMAP output format*. (n.d.). https://colmap.github.io/format.html — *COLMAP text model, quaternions* [S1]
- COLMAP FAQ. (n.d.). *Reconstruct sparse/dense model from known camera poses*. https://colmap.github.io — *run_colmap.sh command sequence, meshing* [S2]
- Schönberger, J. L., & Frahm, J.-M. (2016). *Structure-from-motion revisited*. https://doi.org/10.1007/978-3-319-46487-9_31 — *Photogrammetry engine* [S3]
- Woodham, R. J. (1980). Photometric method for determining surface orientation from multiple images. *Optical Engineering*, *19*(1). https://doi.org/10.1117/12.7972479 — *Photometric stereo* [S4]
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- Kutulakos, K. N., & Seitz, S. M. (2000). A theory of shape by space carving. *International Journal of Computer Vision*. https://doi.org/10.1023/A:1008191222954 — *Space carving* [S7]
- *MVTec HALCON 'photometric_stereo' reference*. (n.d.). https://www.mvtec.com/doc/halcon/13/en/photom — *≥ 3 images, non-coplanar light directions* [S8]
- Fitzgibbon, A. W., Cross, G., & Zisserman, A. (1998). *Automatic 3D model construction for turn-table sequences*. https://doi.org/10.1007/3-540-49437-5_11 — *Turntable (rotating-object) reconstruction* [S9]
- *Meshroom / AliceVision*. (n.d.). https://alicevision.org/ — *Alternative reconstruction* [S10]
- *OpenCV 'solvePnP'*. (n.d.). https://docs.opencv.org/4.13.0/d5/d1f/calib3d_solvePnP.html — *Camera axis convention, pose checks* [S11]
- *OpenCV ChArUco calibration*. (n.d.). https://docs.opencv.org/4.13.0/da/d13/tutorial_aruco_calibration.htm — *Scan-mode intrinsics, scanning mat* [S12]
- Schnieders, D., & Wong, K.-Y. K. (2013). Camera and light calibration from reflections on a sphere. *Computer Vision and Image Understanding*. https://www.sciencedirect.com/science/article/abs/pi — *Chrome-ball light calibration* [S13]
- Yu, Chen, & Da. (2025). Geometric calibration for photometric stereo under near-field point light sources using spheres. *Optics & Laser Technology*, *180*. https://www.sciencedirect.com/science/artic — *Near-light correction (phase 11)* [S14]
- Goldman, D. B., Curless, B., Hertzmann, A., & Seitz, S. M. (2005). *Shape and spatially-varying BRDFs from photometric stereo*. https://doi.org/10.1109/ICCV.2005.219 — *Material on the mesh (phase 13)* [S15]
- *Blender manual: PLY, OBJ, glTF 2.0, formats*. (n.d.). https://docs.blender.org/manual/en/latest/files/import — *Mesh import check* [S16]
- *OpenMVS*. (n.d.). https://github.com/cdcseacave/openMVS — *Optional dense/mesh stage* [S17]
- *RealityScan (formerly RealityCapture)*. (n.d.). https://www.realityscan.com/license — *Optional reconstruction* [S18]
- *QuakeHold! Museum putty*. (n.d.). https://www.sosproducts.com/quakehold-museum-putty-p/64810.htm — *Heavier-object option* [S19]
- *Loctite fun-tak 2 oz*. (n.d.). https://www.staples.com/Loctite-Fun-Tak-2-oz-Mounting-Adhesive-Putty-1270884/product_1480018 — *$3.49 object mount* [S20]

Geometric camera calibration

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- Tsai, R. Y. (1987). A versatile camera calibration technique for high-accuracy 3D machine vision metrology using off-the-shelf TV cameras and lenses. *IEEE Journal of Robotics and Automation*, *3*(4), 323–344. https://doi.org/10.1109/JRA.1987.1087109 — *Classic calibration background (phase 06)* `[V2]`
- Heikkilä, J., & Silvén, O. (1997). A four-step camera calibration procedure with implicit image correction. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 1106–1112. https://doi.org/10.1109/CVPR.1997.609468 — *Distortion model and correction (phase 06)* `[V3]`
- Brown, D. C. (1971). Close-range camera calibration. *Photogrammetric Engineering*, *37*(8), 855–866. https://www.semanticscholar.org/paper/Close-Range-Camera-Calibration-Brown/1150007b62a3c7dac99c2c8f85c63bfab74891af — *Radial/tangential lens distortion (phase 06)* `[V4]`
- Bouguet, J.-Y. (2022). *Camera calibration toolbox for Matlab*. https://doi.org/10.22002/D1.20164 — *Reference calibration workflow (phase 06)* `[V5]`
- Schöps, T., Larsson, V., Pollefeys, M., & Sattler, T. (2020). Why having 10,000 parameters in your camera model is better than twelve. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. https://openaccess.thecvf.com/content_CVPR_2020/html/Schops_Why_Havi — *Calibration accuracy limits (phase 17)* `[V6]`
- Garrido-Jurado, S., Muñoz-Salinas, R., Madrid-Cuevas, F. J., & Marín-Jiménez, M. J. (2014). Automatic generation and detection of highly reliable fiducial markers under occlusion. *Pattern Recognition*, *47*(6), 2280–2292. https://doi.org/10.1016/j.patcog.2014.01.005 — *ArUco markers on ChArUco board (phases 06, 11)* `[V7]`
- Hartley, R., & Zisserman, A. (2004). *Multiple view geometry in computer vision* (2nd ed.). https://doi.org/10.1017/CBO9780511811685 — *Projection, triangulation, bundle adjustment (phases 06, 11)* `[V8]`
- Szeliski, R. (2022). *Computer vision: Algorithms and applications* (2nd ed.). https://doi.org/10.1007/978-3-030-34372-9 — *General CV reference (phases 06, 07, 08, 09, 10, 11, 12, 13)* `[V9]`

Hand-eye and robot-world calibration (AX=XB, AX=YB)

- Tsai, R. Y., & Lenz, R. K. (1989). A new technique for fully autonomous and efficient 3D robotics hand/eye calibration. *IEEE Transactions on Robotics and Automation*, *5*(3), 345–358. https://doi.org/10.1109/70.34770 — *Hand-eye AX=XB baseline (phase 11)* `[V10]`
- Park, F. C., & Martin, B. J. (1994). Robot sensor calibration: Solving AX=XB on the Euclidean group. *IEEE Transactions on Robotics and Automation*, *10*(5), 717–721. https://doi.org/10.1109/70.326576 — *Lie-group hand-eye solver (phase 11)* `[V11]`
- Daniilidis, K. (1999). Hand-eye calibration using dual quaternions. *International Journal of Robotics Research*, *18*(3), 286–298. https://doi.org/10.1177/02783649922066213 — *Simultaneous rotation/translation hand-eye (phase 11)* `[V12]`
- Zhuang, H., Roth, Z. S., & Sudhakar, R. (1994). Simultaneous robot/world and tool/flange calibration by solving homogeneous transformation equations of the form AX=YB. *IEEE Transactions on Robotics and Automation*, *10*(4), 549–554. https://doi.org/10.1109/70.313105 — *AX=YB formulation (phase 11)* `[V13]`
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- Shah, M. (2013). Solving the robot-world/hand-eye calibration problem using the Kronecker product. *Journal of Mechanisms and Robotics*, *5*(3), 031007. https://doi.org/10.1115/1.4024473 — *Closed-form AX=YB (OpenCV SHAH) (phase 11)* `[V15]`
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- Ali, I., Suominen, O., Gotchev, A., & Ruiz Morales, E. (2019). Methods for simultaneous robot-world-hand–eye calibration: A comparative study. *Sensors*, *19*(12), 2837. https://doi.org/10.3390/s19122837 — *Choosing an AX=YB method (phase 11)* [V17]
- *OpenCV, camera calibration and 3D reconstruction ('calibrateHandEye', 'calibrateRobotWorldHandEye')*. (n.d.). https://docs.opencv.org/4.13.0/d9/d0c/group__calib3d.html — *Hand-eye implementation (phase 11)* [V18]

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- Mitsunaga, T., & Nayar, S. K. (1999). Radiometric self calibration. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 374–380. https://doi.org/10.1109/CVPR.1999.78 — *Polynomial response curve (phase 07)* [V19]
- Robertson, M. A., Borman, S., & Stevenson, R. L. (2003). Estimation-theoretic approach to dynamic range enhancement using multiple exposures. *Journal of Electronic Imaging*, *12*(2), 219–228. https://doi.org/10.1117/1.1557695 — *Weighted HDR merge (OpenCV Robertson) (phase 07)* [V20]
- Grossberg, M. D., & Nayar, S. K. (2003). What is the space of camera response functions? *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. https://doi.org/10.1109/TPAMI.2004.88 — *Response-curve prior (EMoR) (phase 07)* [V21]
- Hasinoff, S. W., Durand, F., & Freeman, W. T. (2010). Noise-optimal capture for high dynamic range photography. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 553–560. https://doi.org/10.1109/CVPR.2010.5540167 — *Choosing exposure brackets (phase 07)* [V22]
- Granados, M., Ajdin, B., Wand, M., Theobalt, C., Seidel, H.-P., & Lensch, H. P. A. (2010). Optimal HDR reconstruction with linear digital cameras. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. https://vcai.mpi-inf.mpg.de/projects/opthdr/ — *Noise-weighted HDR merge (phase 07)* [V23]
- Reinhard, E., Ward, G., Pattanaik, S., Debevec, P., Heidrich, W., & Myszkowski, K. (2010). *High dynamic range imaging: Acquisition, display, and image-based lighting* (2nd ed.). https://www.oreilly.com/library/view/high-dynamic-range/9780123749147/ — *HDR formats and theory (phase 07)* [V24]

Photometric stereo

- Hertzmann, A., & Seitz, S. M. (2005). Example-based photometric stereo: Shape reconstruction with general, varying BRDFs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *27*(8), 1254–1264. https://doi.org/10.1109/TPAMI.2005.158 — *Reference-sphere PS for glossy objects (phase 11)* [V25]
- Basri, R., Jacobs, D., & Kemelmacher, I. (2007). Photometric stereo with general, unknown lighting. *International Journal of Computer Vision*, *72*(3), 239–257. https://doi.org/10.1007/s11263-006-8815-7 — *Spherical-harmonic lighting PS (phase 12)* [V26]
- Ackermann, J., & Goesele, M. (2015). A survey of photometric stereo techniques. *Foundations and Trends in Computer Graphics and Vision*, *9*. https://doi.org/10.1561/0600000065 — *PS method overview (phases 11, 12)* [V27]
- Shi, B. et al. (2016). A benchmark dataset and evaluation for non-Lambertian and uncalibrated photometric stereo. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. https://openaccess.thecvf.com/content_cvpr_2016/html/Shi_A_Benchmark_Dataset_CVPR_2 — *PS validation dataset (phase 12)* [V28]
- Mecca, R., Wetzler, A., Bruckstein, A. M., & Kimmel, R. (2014). Near field photometric stereo with point light sources. *SIAM Journal on Imaging Sciences*, *7*(4), 2732–2770. https://doi.org/10.1137/140968100 — *Near-light PS model (phase 12)* [V29]
- Quéau, Y., Durix, B., Wu, T., Cremers, D., Lauze, F., & Durou, J.-D. (2018). LED-based photometric stereo: Modeling, calibration and numerical solution. *Journal of Mathematical Imaging and Vision*, *60*(3), 313–340. https://doi.org/10.1007/s10851-017-0761-1 — *LED anisotropy and falloff model (phases 11, 12)* [V30]

- Logothetis, F., Quéau, Y., Mecca, R., & Cipolla, R. (2017). Semi-calibrated near-light photometric stereo. *SSVM*. https://doi.org/10.1007/978-3-319-58771-4_52 — *Unknown LED intensities (phase 12)* `[V31]`
- Santo, H., Waechter, M., & Matsushita, Y. (2020). Deep near-light photometric stereo for spatially varying reflectances. *European Conference on Computer Vision*. https://doi.org/10.1007/978-3-030-58598-3_9 — *Learned near-light PS option (phase 17)* `[V32]`
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- Ikehata, S. (2018). CNN-PS: CNN-based photometric stereo for general non-convex surfaces. *European Conference on Computer Vision*. https://doi.org/10.1007/978-3-030-01267-0_1 — *Learned PS for glossy objects (phase 17)* `[V34]`
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- Nehab, D., Rusinkiewicz, S., Davis, J., & Ramamoorthi, R. (2005). Efficiently combining positions and normals for precise 3D geometry. *ACM Transactions on Graphics*, *24*(3). https://gfx.cs.princeton.edu/pubs/Nehab_2005_ECP/index.php — *Fuse PS normals with mesh (phase 12)* `[V36]`

Multi-view reconstruction, visual hulls, meshing

- Snavely, N., Seitz, S. M., & Szeliski, R. (2006). Photo tourism: Exploring photo collections in 3D. *ACM Transactions on Graphics*, *25*(3), 835–846. https://doi.org/10.1145/1141911.1141964 — *Incremental SfM background (phase 12)* `[V37]`
- Furukawa, Y., & Ponce, J. (2010). Accurate, dense, and robust multiview stereopsis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *32*(8), 1362–1376. https://doi.org/10.1109/TPAMI.2009.161 — *Patch-based MVS (PMVS) (phase 12)* `[V38]`
- Furukawa, Y., & Hernández, C. (2015). Multi-view stereo: A tutorial. *Foundations and Trends in Computer Graphics and Vision*, *9*. https://doi.org/10.1561/0600000052 — *MVS overview (phase 12)* `[V39]`
- Griwodz, C. et al. (2021). AliceVision meshroom: An open-source 3D reconstruction pipeline. *Proc. ACM MMSys*. https://doi.org/10.1145/3458305.3478443 — *Alternative pipeline citation (phase 12)* `[V40]`
- Matusik, W., Buehler, C., Raskar, R., Gortler, S. J., & McMillan, L. (2000). Image-based visual hulls. *Proc. SIGGRAPH*. https://doi.org/10.1145/344779.344951 — *Silhouette visual hull (phase 12)* `[V41]`
- Hernández Esteban, C., & Schmitt, F. (2004). Silhouette and stereo fusion for 3D object modeling. *Computer Vision and Image Understanding*, *96*(3), 367–392. https://www.sciencedirect.com/science/a — *Hull + stereo fusion on turntable (phase 12)* `[V42]`
- Lorensen, W. E., & Cline, H. E. (1987). Marching cubes: A high resolution 3D surface construction algorithm. *Proc. SIGGRAPH*. https://doi.org/10.1145/37401.37422 — *Voxel grid to mesh (phase 12)* `[V43]`
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- Kazhdan, M., Bolitho, M., & Hoppe, H. (2006). Poisson surface reconstruction. *Eurographics Symposium on Geometry Processing*. https://diglib.eg.org/items/245d6ba1-0209-4365-a4a4-16ce46421620 — *Oriented points to mesh (phase 12)* `[V45]`
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- Zhou, Q.-Y., Park, J., & Koltun, V. (2018). *Open3D: A modern library for 3D data processing*. https://arxiv.org/abs/1801.09847 — *Point cloud/mesh library (phases 12, 13, 14)* `[V47]`

Reflectance capture rigs and polarization

- Debevec, P., Hawkins, T., Tchou, C., Duiker, H.-P., Sarokin, W., & Sagar, M. (2000). Acquiring the reflectance field of a human face. *Proc. SIGGRAPH*. https://doi.org/10.1145/344779.344855 — *Light-stage relighting concept (phases 09, 11)* [V48]
- Ma, W.-C., Hawkins, T., Peers, P., Chabert, C.-F., Weiss, M., & Debevec, P. (2007). Rapid acquisition of specular and diffuse normal maps from polarized spherical gradient illumination. *Eurographics Symposium on Rendering*. https://diglib.eg.org/items/da04b92c-39fe-41d1-a658-a8535521521c — *Polarized diffuse/specular normals (phase 10)* [V49]
- Ghosh, A., Fyffe, G., Tunwattanapong, B., Busch, J., Yu, X., & Debevec, P. (2011). Multiview face capture using polarized spherical gradient illumination. *ACM Transactions on Graphics*, *30*(6). https://doi.org/10.1145/2070781.2024163 — *Multi-view polarized capture (phases 10, 11)* [V50]
- Nayar, S. K., Fang, X.-S., & Boult, T. (1997). Separation of reflection components using color and polarization. *International Journal of Computer Vision*, *21*(3), 163–186. https://doi.org/10.1023/A:1007937815113 — *Diffuse/specular split with polarizer (phase 10)* [V51]
- Wolff, L. B. (1990). Polarization-based material classification from specular reflection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *12*(11), 1059–1071. https://doi.org/10.1109/34.61705 — *Polarization physics background (phase 10)* [V52]

Open-source turntable scanners

- *OpenScan (Raspberry Pi photogrammetry scanner)*. (n.d.). https://openscan.eu/pages/openscan-mini — *DIY scanner prior art (phase 11)* [V53]
- *FabScan, media computing group, RWTH aachen university*. (n.d.). https://hci.rwth-aachen.de/fabscan — *Turntable laser scanner prior art (phase 11)* [V54]
- *BQ ciclop scanner + horus software*. (n.d.). https://github.com/bqlabs/ciclop — *Turntable calibration UX reference (phases 11, 14)* [V55]

Texturing and UV unwrapping

- Lévy, B., Petitjean, S., Ray, N., & Maillot, J. (2002). Least squares conformal maps for automatic texture atlas generation. *ACM Transactions on Graphics*, *21*(3), 362–371. https://doi.org/10.1145/566654.566590 — *LSCM UV unwrapping (phase 13)* [V56]
- *Xatlas, mesh parameterization / UV unwrapping library*. (n.d.). https://github.com/jpcy/xatlas — *Automatic UV atlas (phase 13)* [V57]
- Waechter, M., Moehrle, N., & Goesele, M. (2014). Let there be color! Large-scale texturing of 3D reconstructions. *European Conference on Computer Vision*. https://doi.org/10.1007/978-3-319-10602-1_54 — *View selection for mesh textures (phase 13)* [V58]

Appendix D · Standards, colour, optics, safety

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

Appearance and scatter measurement standards (ASTM)

- ASTM International. (2019). *Standard practice for goniometric optical scatter measurements*. https://store.astm.org/e2387-19.html — *BRDF/BTDF measurement practice (phases 07, 08)* [N1]
- ASTM International. (2003). *Standard practice for angle resolved optical scatter measurements on specular or diffuse surfaces*. https://www.astm.org/DATABASE.CART/WITHDRAWN/E1392.htm — *Historical BRDF practice cited by Labsphere (phase 07)* [N2]
- ASTM International. (2005). *Standard practice for goniophotometry of objects and materials*. https://store.astm.org/e0167-96.html — *Goniophotometer geometry conventions (phase 06)* [N3]
- ASTM International. (n.d.). *Standard test method for specular gloss*. https://store.astm.org/standards/d52 — *20/60/85° gloss cross-check (phase 07)* [N4]
- ASTM International. (2023b). *Standard test methods for measurement of gloss of high-gloss surfaces by abridged goniophotometry*. https://store.astm.org/standards/e430 — *Haze/distinctness-of-image near specular (phases 08, 17)* [N5]
- ASTM International. (2025a). *Standard terminology of appearance*. https://doi.org/10.1520/E0284-25 — *Docs glossary terms (phase 00)* [N6]
- ASTM International. (2026). *Standard practice for specifying the geometry of multiangle spectrophotometers*. https://store.astm.org/standards/e2175 — *Reporting light/camera angle geometry (phase 08)* [N7]
- ASTM International. (2025b). *Standard test method for multiangle color measurement of metal flake pigmented materials*. https://doi.org/10.1520/E2194-14R25 — *Aspecular angles for metallic samples (phases 08, 09)* [N8]
- ASTM International. (2023a). *Standard practice for obtaining spectrometric data for object-color evaluation*. https://store.astm.org/standards/e1164 — *Spectral data and calibration practice (phase 10)* [N9]
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Appendix E · Motion control, electronics, open hardware

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

Stepper drivers and microcontrollers

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Interface electronics and position sensors

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Camera interface

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- USB Implementers Forum. (2012). *Video class v1.5 document set*. https://www.usb.org/document-library/video-class-v15-document-set — *UVC exposure/white-balance controls (phase 06)* [H14]
- Linux kernel documentation. (n.d.). *The linux USB video class (UVC) driver*. https://www.kernel.org/doc/ht api/media/drivers/uvcvideo.html — *uvcvideo controls, extension units (phases 06, 14)* [H15]

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Low-cost open instruments (HardwareX and related)

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Design tools and project publishing

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- GitHub Docs. (n.d.-b). *REST API endpoints for milestones*. https://docs.github.com/en/rest/issues/mileston — *Phase milestones and issue scripts (phase 00)* [H44]
- GitHub Docs. (n.d.-a). *Configuring a publishing source for your GitHub pages site*. https://docs.github.com/en/pages/getting-started-with-github-pages/configuring-a-publishing-source-for-your-github-pages-site — *Documentation site deployment (phase 16)* [H45]

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- Harris, C. R., Millman, K. J., Walt, S. J. van der, et al. (2020). Array programming with NumPy. *Nature*, *585*, 357–362. https://doi.org/10.1038/s41586-020-2649-2 — *Image arrays, HDR, geometry (phase 07)* [H47]
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